

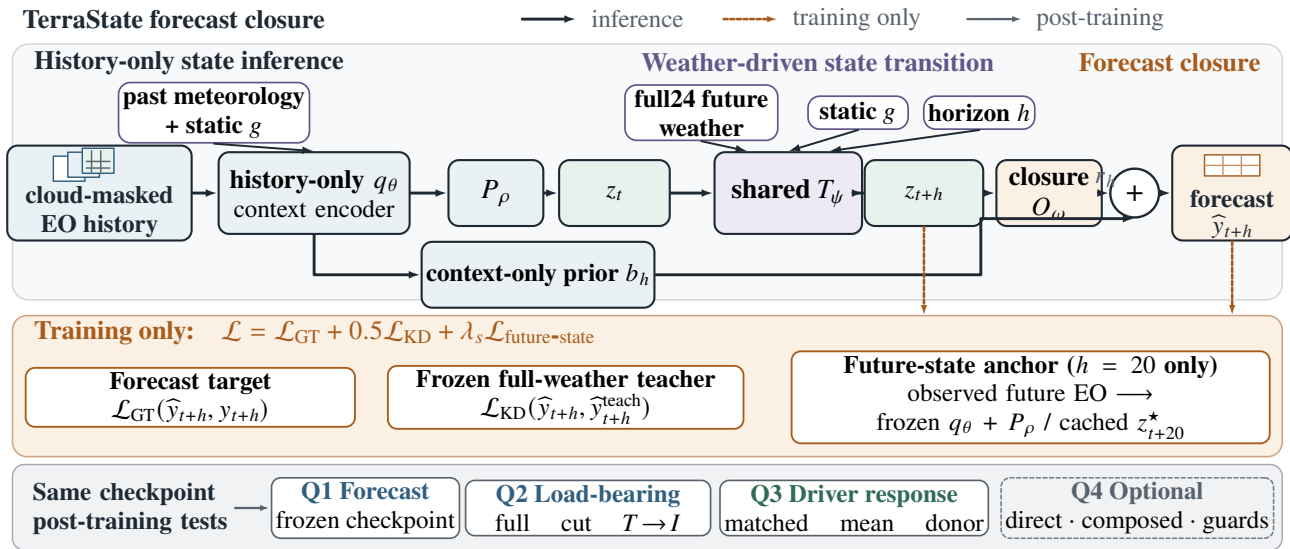


Figure 1: TerraState’s solid inference path is history-only: cloud-masked EO history, retained past meteorology, and static geography enter q_θ , which emits the context-only prior b_h and, through P_ρ , the spatial state z_t . Full24 future weather, static geography, and horizon h condition the shared T_ψ ; the state contribution $O_\omega(z_{t+h})$ is then added explicitly to b_h . The orange band contains only training supervision: the ground-truth forecast loss, distillation from a frozen full-weather teacher, and the $h = 20$ future-state anchor from a frozen future-observation encoder/cache. Neither teacher nor observed future enters inference. The bottom strip summarizes same-checkpoint post-training evidence: Q1 forecast skill, Q2 closure and transition cuts, Q3 matched weather controls, and optional Q4 direct/composed queries under endpoint and non-collapse guards.